

Lecture 7 Amortized Analysis

Recall

Hashing with Chaining.

Recall:

Hashing with Chaining:

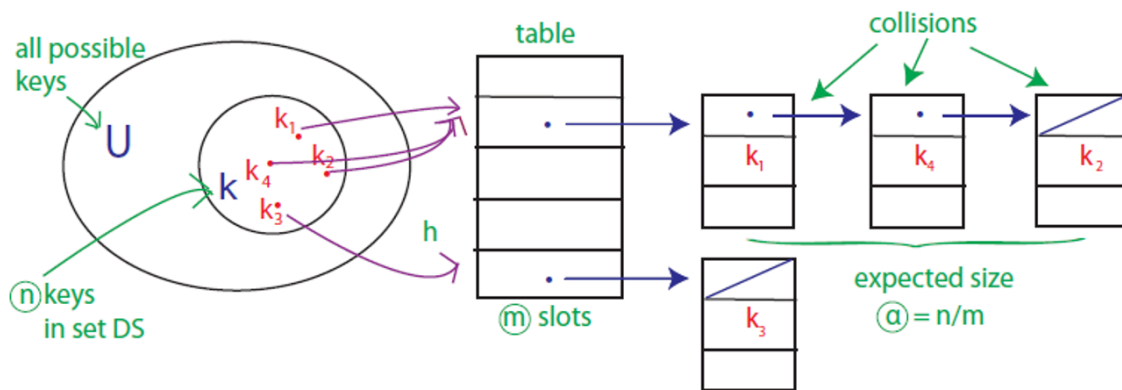


Figure 1: Hashing with Chaining

Expected cost (insert/delete/search): $\Theta(1 + \alpha)$, assuming simple uniform hashing *OR* universal hashing & hash function h takes $O(1)$ time.

Rehashing

m too small causes slow and m too large causes wasteful. Thus $m = \Theta(n)$ at all times. For each rebuilding, we must insert items in old table with n elements into new table with $m = \Theta(n)$ slots. Each rebuilding costs $\Theta(n + m) = \Theta(n)$

Two Strategies to grow

When n reaches m , we can

- $m_+ = 1$. Rebuild every step and copy i elements. n inserts cost $\Theta(1 + 2 + \dots + n) = \Theta(n^2)$.
- $m_* = 2$. Rebuild at insertion 2^i . n inserts cost $\Theta(1 + 2 + 4 + \dots + n) = \Theta(n)$.

Analysis

Let c_i be the cost of i th insertion. When $i - 1$ is the power of 2, $c_i = i$; otherwise $c_i = 1$. Total cost of copy is $1 + 2 + 4 + \dots + n = \Theta(n)$, total cost of inserts is n . Total cost is

$$n + \Theta(n) = \Theta(n).$$

Cannot use worst cost of one operation multiply the number of operations. The worst cost does not occurs each time.

Thus, average cost per insert is $\frac{\Theta(n)}{n} = \Theta(1)$.

Amortized Analysis

Analyze a seq. of operations to show that average cost per operation is small even though one operation may be expensive.

It is average performance in worst case.

Three types:

- Aggregate(just saw)
- Accounting
- Potential

Accounting Method

- charge i th operation a fictitious amortized cost $\hat{C}_i$
- Fee is consumed to perform operation
- Unused amount stored in bank for use by later operations
- bank balance must not go negative

we have

$$\sum_{i=1}^n C_i \leq \sum_{i=1}^n \hat{C}_i, \forall n$$

for example,

插入序号	表大小	实际代价 C_i	摊还代价 $\hat{C}_i$	银行余额
1	1	1	3	2
2	2	2	3	3
3	4	3	3	3
4	4	1	3	5
5	8	5	3	3
...	...	...	...	...

Potential Method

Frame work

- Start with data structure D_0
- Operation i transforms D_{i-1} to D_i

Define potential function Φ maps D_i into a real number. and $\Phi(D_i) \geq 0, \forall i$.

Amortized cost $\hat{C}_i$ is $\hat{C}_i = C_i + \Phi(D_i) - \Phi(D_{i-1})$.

Potential difference $\Delta\Phi_i = \Phi(D_i) - \Phi(D_{i-1})$. If $\Delta\Phi_i > 0$, operation i stores work in data structure for later. Otherwise, it delivers up stored work to help pay for operation i .

Total amortized cost of n operations is

$$\sum_{i=1}^n \hat{c}_i = \sum_{i=1}^n c_i + \Phi(D_n) - \Phi(D_0) \geq \sum_{i=1}^n c_i$$

Define potential function onto rare resources.